CloudCampus-AI

AWS Cost Optimizer and Resource Monitoring Platform

Project Type	Full-Stack Cloud Computing and Machine Learning Project
Domain	Cloud Computing, AWS, Cost Optimization, Resource Monitoring
Frontend	React, TypeScript, Vite, Recharts
Backend	Python, FastAPI, SQLAlchemy, Alembic
Database	MySQL 8
Cloud Services	AWS EC2, CloudWatch, Cost Explorer, STS, IAM
Machine Learning	Isolation Forest for anomaly detection and Linear Regression for cost forecasting

Synopsis submitted for academic project documentation and viva presentation.

1. Abstract

CloudCampus-AI is a full-stack cloud cost optimization and resource monitoring platform designed to help users monitor AWS resources, analyze cloud expenditure, detect unusual billing patterns, and forecast future cloud costs. The system integrates AWS services with a FastAPI backend, MySQL database, React dashboard, and machine learning modules to provide a centralized and beginner-friendly view of cloud usage.

The project addresses a practical problem in cloud computing: resources are often created for testing, learning, or deployment and then left idle, causing unnecessary billing. CloudCampus-AI helps identify such inefficiencies through resource synchronization, CloudWatch metrics, Cost Explorer data, optimization rules, anomaly detection, and forecasting.

2. Problem Statement

Cloud platforms provide powerful services, but beginners and small teams often struggle to track resource usage and spending across multiple AWS consoles. EC2 instances may remain running even when they are underutilized, cost spikes may go unnoticed, and future expenses may be difficult to estimate. Existing AWS tools are useful, but they can be complex for students and early-stage users who need a simple, project-oriented dashboard.

The problem is to design and develop a centralized platform that connects with AWS, collects resource and billing information, analyzes usage patterns, detects anomalies, forecasts cost, and provides actionable recommendations for optimization.

3. Objectives

- To build a centralized dashboard for AWS resource monitoring and cost analysis.
- To fetch EC2 resource information using AWS SDK integration.
- To collect CPU utilization and performance metrics through AWS CloudWatch.
- To retrieve daily billing information using AWS Cost Explorer.
- To detect abnormal cloud cost behavior using machine learning.
- To forecast future AWS spending from historical cost data.
- To generate optimization recommendations for underutilized cloud resources.
- To implement secure authentication and role-based access for Admin, Faculty, and Student users.

4. Scope of the Project

The current scope focuses on AWS EC2 resource monitoring, CloudWatch metric analysis, AWS cost data collection, anomaly detection, cost forecasting, and recommendation lifecycle management. Admin users can synchronize resources, run optimization scans, collect cost data, detect anomalies, generate forecasts, and update recommendation status as OPEN, RESOLVED, or IGNORED.

The system can be extended in future versions to include additional AWS services such as S3, RDS, Lambda, EKS, ECS, and automated notification workflows.

5. Technologies Used

Layer	Technologies	Purpose
Frontend	React, TypeScript, Vite, React Router, Recharts, CSS	Dashboard UI, charts, routing, and user interaction
Backend	Python, FastAPI, Uvicorn	REST API development and server execution
Database	MySQL 8, SQLAlchemy, Alembic	Data storage, ORM mapping, and schema migrations
Cloud Integration	AWS Boto3 SDK	Programmatic access to AWS services
Machine Learning	Scikit-learn	Anomaly detection and cost forecasting
Testing and Quality	Pytest, FastAPI TestClient, Oxlint, TypeScript build	Smoke testing, linting, and production build validation
Version Control	Git	Source code tracking and project management

6. Cloud Services Used

AWS Service	Role in the Project
Amazon EC2	Fetches and monitors virtual machine instances, their state, and resource information.
Amazon CloudWatch	Collects CPU utilization and performance metrics for EC2 monitoring.
AWS Cost Explorer	Retrieves historical daily cost and billing data for analysis and forecasting.
AWS STS	Verifies AWS account identity and account ID.
AWS IAM	Provides permission-based access using AWS profiles and policies.

7. System Architecture

CloudCampus-AI follows a layered architecture. The React frontend communicates with the FastAPI backend through REST APIs. The backend handles authentication, authorization, AWS integration, database operations, optimization logic, and machine learning processing. MySQL stores users, cloud resources, cost records, anomalies, metric snapshots, and recommendations.

Component	Responsibility
User Interface	Provides dashboards, charts, forms, filters, and role-based views.
API Layer	Handles requests, authentication, RBAC, validation, and response formatting.
AWS Integration Layer	Uses Boto3 to collect EC2, CloudWatch, STS, and Cost Explorer data.

Component	Responsibility
Database Layer	Stores persistent application data in MySQL using SQLAlchemy models.
ML Layer	Runs Isolation Forest for anomaly detection and Linear Regression for forecasting.
Optimization Engine	Analyzes usage metrics and creates cost-saving recommendations.

8. Major Modules

Authentication and RBAC Module

Manages user registration, login, JWT token generation, password security, and role-based access for Admin, Faculty, and Student users.

AWS Resource Module

Synchronizes EC2 instance data such as instance ID, state, region, and resource metadata from AWS.

Monitoring Module

Collects CPU utilization and performance-related metrics from Amazon CloudWatch.

Cost Analysis Module

Retrieves and stores daily AWS cost data using Cost Explorer for reporting and analysis.

Anomaly Detection Module

Uses Isolation Forest to identify unusual spending patterns in cloud cost data.

Cost Forecasting Module

Uses Linear Regression to predict future cloud expenditure from historical cost records.

Optimization Recommendation Module

Analyzes resource utilization and generates recommendations for underutilized cloud resources. Admin users can mark recommendations as OPEN, RESOLVED, or IGNORED.

9. Methodology

- The user logs in through the frontend, and the backend verifies identity using JWT authentication.
- The backend checks role permissions before allowing access to protected cloud and dashboard features.
- AWS resource data is collected using Boto3 and stored in MySQL.
- CloudWatch metrics are fetched to analyze EC2 utilization.
- Cost Explorer data is collected and saved for historical analysis.
- Machine learning models process cost data to detect anomalies and forecast upcoming expenses.
- The optimization engine evaluates usage patterns and creates actionable recommendations.

- The frontend displays the result through dashboards, charts, resource tables, and recommendation workflows.

10. Expected Outcome

The expected outcome is a working full-stack platform that helps users understand AWS resource usage, monitor cloud cost, detect unusual spending behavior, forecast future expenses, and identify optimization opportunities. The project demonstrates practical knowledge of AWS cloud services, secure API development, database management, frontend engineering, and applied machine learning.

11. Future Enhancements

- Add support for AWS S3, RDS, Lambda, ECS, and EKS monitoring.
- Introduce email or dashboard alerts for cost spikes and budget limits.
- Add automated shutdown suggestions or controlled automation for idle resources.
- Containerize the application using Docker and Docker Compose.
- Add CI/CD pipeline support for testing and deployment.
- Extend the platform to support Azure and Google Cloud in future versions.

12. Conclusion

CloudCampus-AI is a practical cloud computing project that combines AWS integration, full-stack web development, database design, authentication, cost analysis, and machine learning. It solves a real-world problem by helping users monitor cloud resources and control unnecessary expenses. The system is especially useful for students, beginners, and small teams who want a simplified platform for cloud cost visibility and optimization.

Viva Summary

CloudCampus-AI is a FastAPI and React based AWS cost optimization platform that monitors EC2 resources, collects CloudWatch and Cost Explorer data, detects anomalies using Isolation Forest, forecasts future cloud cost using Linear Regression, and provides optimization recommendations through a role-based dashboard.